

1977

447 SECONDS

THE LENGTH OF TIME THE GOSSAMER CONDOR FLEW TO WIN THE KREMER PRIZE

The Gossamer Condor was the first human-powered aircraft to achieve sustained flight.

IN FEBRUARY, a team of oceanographers and geologists set out to explore the Galápagos Ridge in the East Pacific Ocean, searching for **hydrothermal vents**—cracks in the ocean floor where seawater meets hot magma. They found that the mineral-rich water gushing out of the hydrothermal vents supported a rich and diverse community of organisms never seen before. Later in the year, American paleobotanist **Elso Barghoorn** (1915–84) and his Ph.D. student Andrew Knoll discovered 3.4 billion-year-old fossils of simple single-celled organisms in rocks

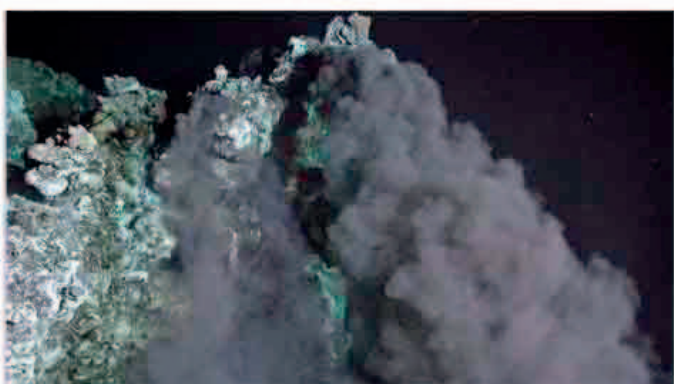
in South Africa. Their find pushed back the **date of the earliest known life** by several hundred million years.

In the same year, the science of **genomics**—analysis of the sequence of nucleotide bases (genetic material) along the length of an organism's DNA—took two major steps forward. First, in February, British biochemist **Frederick Sanger** determined the sequence of the 5,000 or so bases in the genome of a simple virus. Then, in December, Sanger published details of a new, rapid method of **genome sequencing**. The

Sanger method remained the basis of genome sequencing until automated techniques became available in the early 2000s.

In August, **Jerry Ehman**, an American astronomer, detected a remarkably strong and consistent **radio signal** that seemed to have come from an **extraterrestrial source**. The signal's strength, frequency, and consistency suggested that it was broadcast on purpose by intelligent extraterrestrial beings. Ehman used Ohio State University's "Big Ear" radio telescope as part of the **Search for Extraterrestrial Intelligence (SETI)** project. The signal has never been detected again.

Also in August, the **Gossamer Condor** became the first **human-powered aircraft** to achieve sustained, controlled flight, completing a figure of eight around two markers set 0.4 mile (0.8 km) apart, at Minter Field, California. It had a mass of just 70 lb (32 kg) and was driven by pedal power. The aircraft's designer, Paul MacCready, was awarded the **Kremer Prize** (£50,000), which was set up in 1959 by British industrialist Henry Kremer to reward innovations in human-powered flight.



HYDROTHERMAL VENTS

A common feature around mid-ocean ridges, hydrothermal vents are fissures in the ocean floor from which chemically rich water emerges. They can either be hot "black smokers," which form rocky, chimneylike structures, or cool "white smokers." Some biologists believe white smokers to be the site of the origin of life.

1978

“IN A WAY, I’M PROUD THAT WHAT I WAS HAS HELPED OTHER PEOPLE HAVE CHILDREN.”

Louise Brown, first baby conceived through IVF, 1988



MARY LEAKEY (1913–96)

Born in London, Mary Nicol married archaeologist and anthropologist Louis Leakey. The couple spent many years in Africa, inspired by the idea that human origins lay in that continent. In 1959, Mary found the 1.75 million-year-old remains of a hominid in Tanzania. Her find contributed to later understanding of human evolution.

THE US AIR FORCE LAUNCHED the first of 24 NAVSTAR (Navigation System using Timing And Ranging) satellites for its **Global Positioning System (GPS)** in February. The satellites acted as orbiting radio beacons. Each one carried an accurate atomic clock and broadcast constant signals announcing its position and the exact time. Ground-based receivers could detect signals from at least four satellites from any point on

Earth. A receiver worked out its exact location by triangulating the signals from the satellites. In 1983, US President Ronald Reagan announced that the system would be available for public use once the first group of satellites was complete.

A second group of satellites, the first of which was launched in 1989, enhanced accuracy.

Also in February, a team led by British paleontologist Mary Leakey revealed a set of **prehistoric footprints** that were made by bipedal hominids an estimated **3.4 million years ago**. The 80 ft- (24 m-) long set of footprints, discovered at **Laetoli**, in Tanzania, were made by three individuals walking in volcanic ash. Light rainfall soon cemented the footprints, which were then covered in another layer of ash and preserved until erosion revealed them to Leakey and her team. The oldest hominid footprints known before the discovery at Laetoli were made by Neanderthals just 80,000 years ago.

In July, the first human baby conceived outside a woman's body was born. Louise Brown was conceived through **in vitro fertilization (IVF)**—a technique that was the result of nearly a decade of work by British embryologist Robert Edwards and surgeon Patrick Steptoe.

